

EMBEDDED SYSTEM AND DESIGN

EXPERIMENT NO. 4

KETTLE

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DISCUSSION

A kettle is a device primarily developed to efficiently and conveniently heat water for various purposes. From traditional models that should be put on a stove to more modern and electric types, kettles have been part of people's necessities for a long time. For this experiment, the evolution of the kettle will be demonstrated by recreating a kettle's automatic feature of turning off after reaching the water's boiling point. The demonstration will mainly use an Arduino UNO, a humidity sensor, a sound sensor, and a servo motor.

The experiment mainly focuses on the automation system of a kettle, particularly the turning off of the heater when the contained water reaches boiling point. Hence, an automatic electric kettle was not considered to be utilized in this experiment to show the turning off of the system. The logic behind the experiment is as

follows: the humidity and sound sensor continue to detect the corresponding inputs; for the humidity, steam is the primary source of percentage, while the whistle of the kettle will be the source for the sound sensor; when both sensors are triggered, the servo motor will then move, specifically rotate, to turn off the cooker. For the effect of steam on humidity, the room percentage of humidity is 30%, while a room-with-steam percentage of humidity reaches 81%. However, if a room is air-conditioned, the percentage lowers to 60%. In terms of the code, two libraries are utilized so that the humidity sensor and the servo motor will work: the "dht.h" library and the "Servo.h" library. Both are external libraries that made the coding more convenient because pin configurations and if-else statements are the only necessities to be done with these libraries. The if-else statements employed show that if the humidity percentage reaches 60% AND if the sound sensor detects a whistling sound, which is LOW,

similar to a switch button, then the servo motor will move with an angle from 0 to 180 degrees.

CONCLUSION

In conclusion, a real kettle showed the automated portion of turning off the heater. The servo motor's motion proves that the water's boiling point was indicated and appropriately recorded by checking the humidity percentage and sound a short time after the kettle's whistling and steaming actions. As usual, the lessons in microprocessor, logic circuits, and switching theory were applied through pin configuration, triggering of sensors, and proving a kettle's process with an embedded system. Hence, a real kettle, particularly an electric kettle's automated feature, was produced and demonstrated successfully through an embedded system.